INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CHEMISTRY 9729/03

Paper 3 Free Response

10 September 2018

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

This document consists of **13** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

1 This question is about the chemistry of ethene and its derivatives.

(a) Ethene is the starting material to form ethanedioic acid.

(i) Suggest the synthetic route for the formation of ethanedioic acid from ethene. [2]

(ii) 0.200 mol of ethene is stored in a 20.0 dm^3 flask with 0.800 mol of ethane at 127 °C.

Calculate the total pressure in the flask. Hence or otherwise, calculate the partial pressure of ethene in the flask. [2]

(iii) The total pressure that you calculated in **(a)(ii)** is different from the actual pressure exerted. Suggest an explanation for the difference. [1]

(iv) Ethene reacts with hydrogen in the presence of nickel catalyst to form ethane.

Explain why nickel can be used as a catalyst in this reaction. [2]

(v) Outline the mode of action of the catalyst in this reaction. [2]

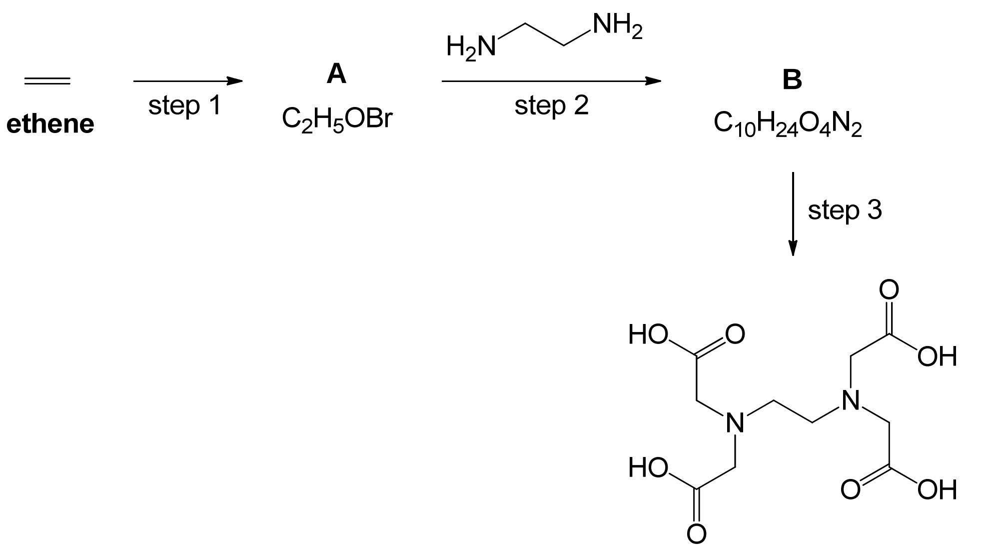
(b) Dissolving 4.82×10^{-5} mol calcium ethanedioate, CaC_2O_4, in 1 dm^3 of water forms a saturated solution.

(i) Write an expression for the solubility product of calcium ethanedioate and state its units. [1]

(ii) Calculate the solubility product of calcium ethanedioate. [1]

(iii) 50.0 cm^3 of 0.100 mol dm^{-3} $CaCl_2$ and 50.0 cm^3 of 0.300 mol dm^{-3} $Na_2C_2O_4$ are mixed together. Determine if CaC_2O_4 will be precipitated out. [2]

(c) Synthesis of the unionised form of EDTA (H_4Y) can be carried out in the laboratory using ethene as the starting material.

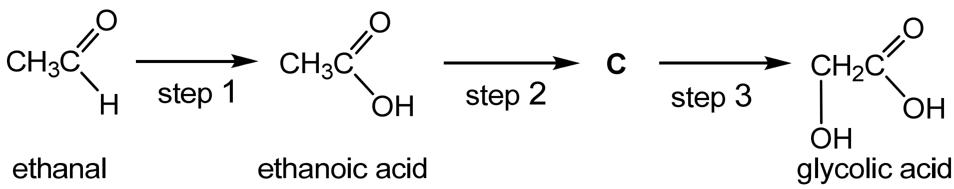


unionised form of EDTA (H_4Y)

(i) Draw the structures of **A** and **B**. [2]

(ii) Suggest appropriate reagents and conditions for step 3. [1]

(d) Ethene can be used to form ethanal. Ethanal in turn is used to synthesize glycolic acid via the reaction shown below.



(i) Identify the intermediate **C** and state the reagent and condition for step 3. [2]

(ii) Explain the difference in acidity between glycolic acid and ethanoic acid. [2]

(iii) Two molecules of glycolic acid can react with one another under suitable conditions to form a neutral compound with the loss of two water molecules.

Suggest a possible structure for the compound formed. [1]

(iv) Suggest a simple chemical test to distinguish between ethanoic acid and glycolic acid. [2]

[Total: 23]

2 Iron is the fourth most common element in the Earth's crust. It is a d-block element which is known to exhibit different characteristics from the s-block elements. Since ancient times, iron has been widely employed in a variety of applications.

(a) **(i)** One well-known property of iron and its compounds is the ability to catalyse reactions. For example, aqueous iron(II) chloride can be used to catalyse the reaction between I^- and $S_2O_8^{2-}$, to form I_2 and SO_4^{2-} ions.

Using relevant $E^\ominus$ values from the *Data Booklet*, explain why iron(II) chloride can be used as a catalyst for this reaction.

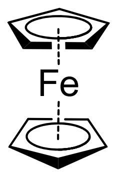
[2]

(ii) A sample of iron was vapourised, ionised and passed through an electric field. Analysis of deflection occurring at the electric field region revealed that a sample of $^{32}S^{2-}$ ions would be deflected by +20° towards the positive potential.

What is the angle, and direction of deflection for a sample of $^{56}Fe^{3+}$ ions passing through the same electric field?

[2]

(b) Ferrocene, $Fe(C_5H_5)_2$, is an orange solid which is known to exhibit anti-cancer activity. In this complex, $C_5H_5^-$ is the ligand and it donates π electrons from the ring to the vacant 3d orbital of Fe. The structure of ferrocene is given below.



ferrocene

(i) State the oxidation number of Fe in ferrocene.

[1]

(ii) Suggest why ferrocene is a coloured complex.

[2]

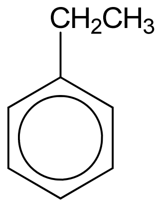
(iii) Light of a longer wavelength is lower in energy than light of a shorter wavelength. The following table shows the spectral colours and the corresponding wavelengths.

Colour	Wavelength / nm
Violet	380 – 450
Blue	450 – 495
Green	495 – 570
Yellow	570 – 590
Orange	590 – 620
Red	620 – 750

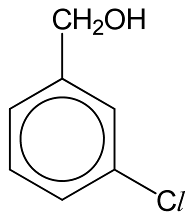
Given that aqueous Fe^{2+} ion is green in colour, suggest and explain if water causes a larger split between the two groups of 3d orbitals as compared to $C_5H_5^-$.

[2]

(c) Ethylbenzene is used to synthesize (3-chlorophenyl)methanol which is used as a general solvent for inks, paints, lacquers, epoxy resin coatings and as a degreasing agent.



ethylbenzene



(3-chlorophenyl)methanol

Starting with ethylbenzene, outline a three-step reaction scheme to obtain (3-chlorophenyl)methanol. Your answer should include clearly the reagents and conditions in each step, and the structures of all intermediates formed.

[3]

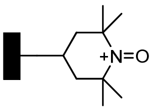
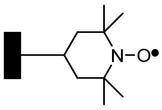
(d) **D** is an achiral organic compound with the molecular formula C_2H_7NO. It can be formed from the reaction between a primary amide and lithium aluminium hydride. In the presence of a suitable catalyst, 1 mole of **D** reacts with 1 mole of benzoic acid to form **E**, $C_9H_{11}NO_2$. However, 1 mole of **D** requires 2 moles of benzoyl chloride to react completely to form **F**, $C_{16}H_{15}NO_3$ and copious white fumes. 2 moles of **D** can also react with gaseous PCl_5 to form a cyclic **G**, $C_4H_{10}N_2$, which contains a 6-membered ring.

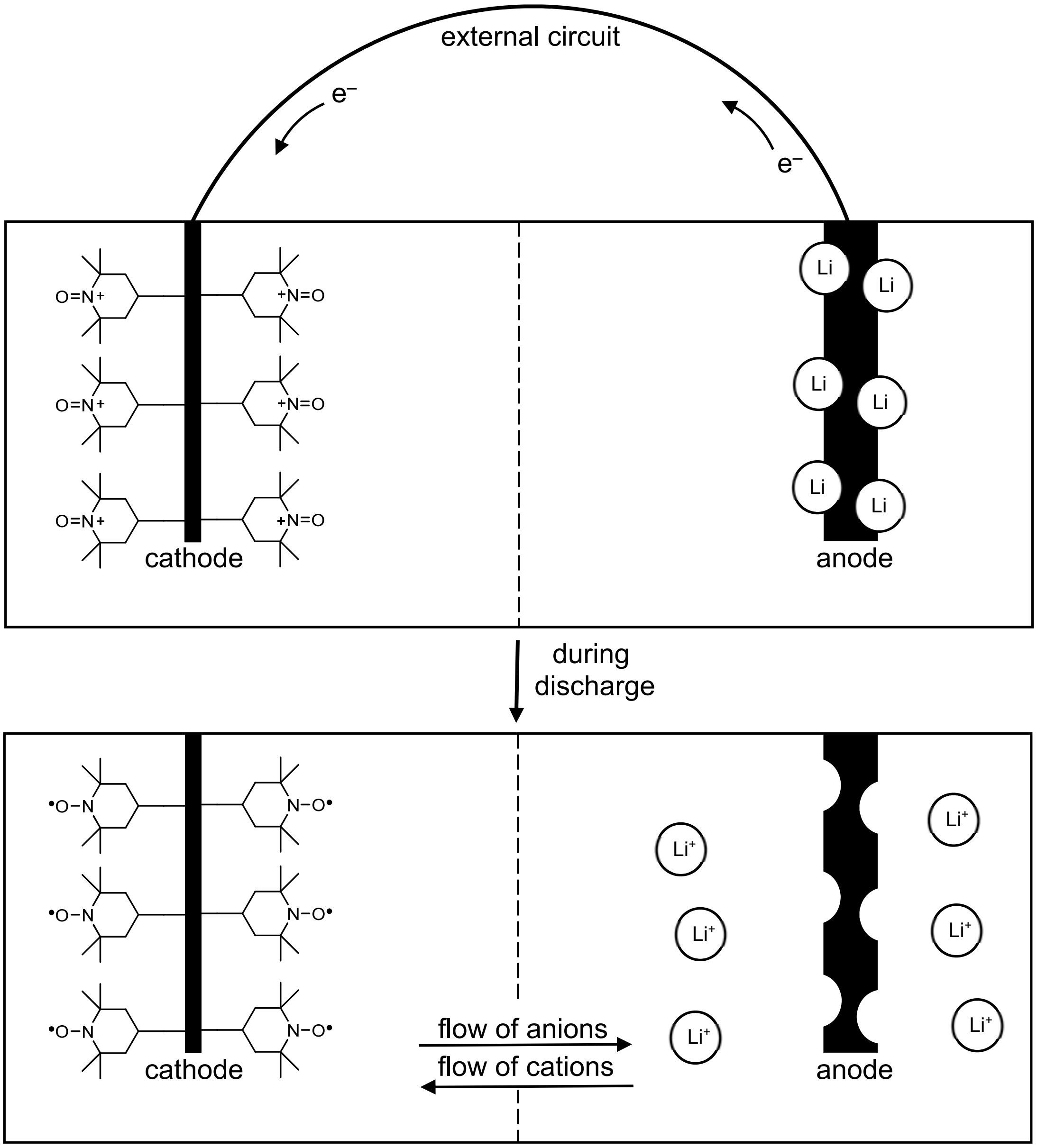
Deduce the structures of **D**, **E**, **F** and **G** and explain the reactions described.

[7]

[Total: 19]

3 An organic radical battery (ORB) is a relatively new type of battery which uses flexible plastics, to provide electrical power. One type of hybrid ORB/Li-ion battery consists of: a cathode made from solid organic polymer containing oxoammonium ion formed from (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl (TEMPO) nitroxide radicals; an anode made from graphite with Li atoms inserted between the graphite layers; and an electrolyte of $LiPF_6$ dissolved in organic solvent.

During discharge, Li atoms give up electrons at the anode to become Li^+ ions. The electrons travel round the external circuit, and are picked up by the cathode. The anions and cations in the electrolyte move to the anode and cathode respectively. This is illustrated in the following diagram in which  and  are simplified representations of the polymer containing oxoammonium ion and TEMPO nitroxide radicals respectively.



(a) **(i)** Graphite is often mixed in the polymer used for making the cathode electrode.

Suggest a reason for this. [1]

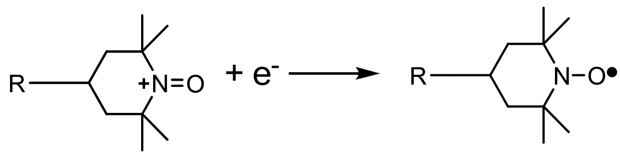
(ii) Calculate the oxidation number of N in the cathode **before** discharge. [1]

(iii) The E_{cell} generated by the hybrid ORB/Li-ion battery under standard conditions is 2.17 V.

Use relevant $E^\ominus$ value from the *Data Booklet* to calculate the electrode potential generated by the cathode half-cell.

[1]

(iv) During discharge, the following reaction occurs at the cathode.



where R represents the organic polymer cathode.

Write an equation for the overall process that occurs during discharge.

[1]

(v) Draw the dot-and-cross diagram of the PF_6^- ion and state its shape.

[2]

(vi) Suggest whether $LiPF_6$ or LiF has a lower melting point. Explain your answer.

[2]

(b) The hybrid ORB/Li-ion battery is a secondary battery, i.e., it is rechargeable.

During charging, 1.22 g of Li is regenerated from Li^+ ions at the cathode.

(i) Calculate the amount of electrons required to form 1.22 g of Li. [1]

Besides the generation of Li, there is a competing side-reaction that occurs at the cathode.

In this side-reaction, ethylene carbonate, $C_3H_4O_3$ undergoes reduction in the presence of Li^+ ions to form ethene and lithium carbonate.

(ii) Write the half-equation for the side-reaction occurring at the cathode.

[1]

A current of 5.0 A is supplied over 2 hours during charging.

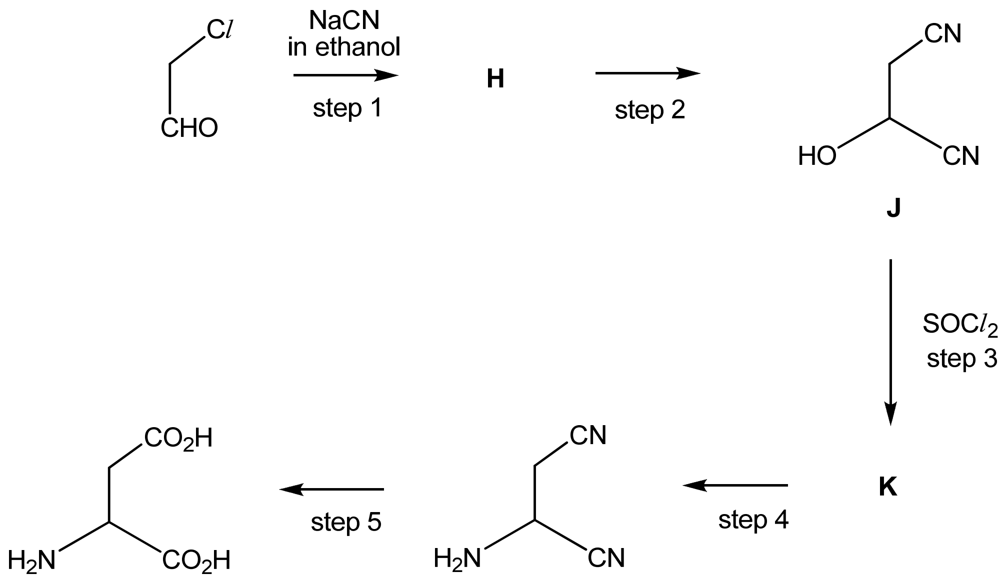
(iii) Use the information given and your answer in **(b)(i)** to calculate the amount of electrons consumed by the side-reaction.

[1]

(iv) Suggest why the battery needs to be replaced after about 1000 charge-discharge cycles.

[1]

(c) Chloroethanal is the starting material in the synthesis of aspartic acid.



(i) Suggest structures for the intermediates **H** and **K**.

[2]

(ii) Suggest reagents and conditions for step 2 and for step 4.

[2]

(iii) The reaction in step 2 produces sample **J**, which does not show optical activity. Explain the observation.

[2]

[Total: 18]

Section B

Answer **one** question from this section

4 **(a)** For many compounds the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy changes of combustion can be used.

The enthalpy change of combustion can be found by a calorimetry experiment in which the heat energy given off during combustion is used to heat a known mass of water and the temperature change recorded.

(i) Define the term *standard enthalpy change of combustion.*

[1]

(ii) Write the equation for the complete combustion of ethanol, CH_3CH_2OH.

[1]

In an experiment to determine the enthalpy change of combustion of ethanol, 0.23 g of ethanol was burned and the heat given off raised the temperature of 100 g of water by 16.3 °C.

(iii) Calculate the heat energy change during the combustion of ethanol.

[1]

(iv) Hence, calculate the enthalpy change on burning 1 mole of ethanol.

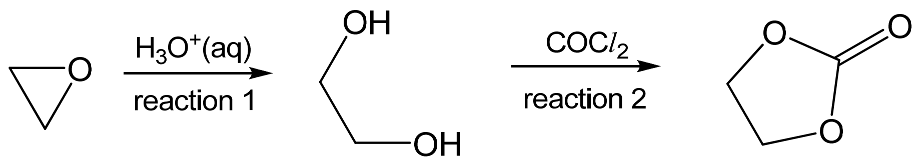
[2]

(v) Suggest one reason why the value for the enthalpy change of combustion of ethanol determined by a simple laboratory calorimetry experiment is likely to be lower than the true value.

[1]

(b) Epoxides are cyclic ethers commonly used in organic reactions.

Ethylene carbonate can be prepared from an epoxide, oxirane by the following reactions.

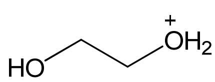


oxirane ethylene carbonate

(i) Suggest the type of reaction for reaction 2.

[1]

(ii) Reaction 1 is an acid-catalysed reaction that proceeds via a three-step mechanism:

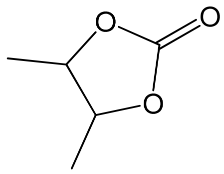
1) Protonation of oxirane by H_3O^+.
2) Ring opening of protonated oxirane due to nucleophilic attack by H_2O to yield an oxonium ion, 

.
3) Deprotonation of the oxonium ion to yield the product, with the regeneration of H_3O^+.

Suggest the mechanism for Reaction 1.

[3]

Compound **L** can be synthesised from an epoxide in a similar manner as ethylene carbonate.



Compound **L**

(iii) Draw the structure of the epoxide used for synthesising compound **L**.

[1]

(iv) Suggest why compound **L** is able to exhibit cis-trans isomerism.

[1]

(c) A number of isomers with the formula $Fe(H_2O)_6Cl_3$ exist. Their general formula is $[Fe(H_2O)_{6-n}Cl_n]Cl_{3-n}.nH_2O$.

Each isomer contains a six co-ordinated Fe(III) ion in an octahedral complex. Water molecules not directly bonded with the Fe atom are held in the crystal lattice as water of crystallisation.

(i) Similar to organic compounds, octahedral complexes can also exhibit stereoisomerism depending on the orientation of the ligands.

One such example will be the iron complex when *n* is 2. It can exist in two isomeric forms where only one of them has a dipole moment.

Name the type of isomerism shown by the complex.

[1]

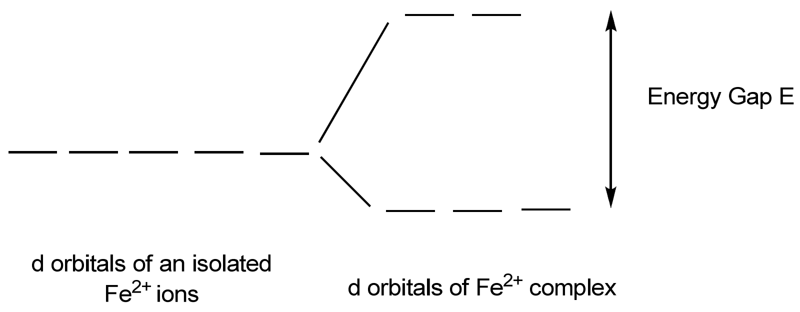
(ii) Draw the structures of the two isomeric forms of the complex.

[2]

(iii) State which isomer has a dipole moment. Explain your answer.

[2]

(iv) The following diagram shows how the d-orbitals are split in an octahedral environment.



When the H_2O ligand is changed to a C*l* ligand, the Fe^{2+} ion changes the electronic configuration from a 'high spin' to a 'low spin' state.

In a 'high spin' state, the electrons occupy all the d-orbitals singly, before starting to pair up in the lower energy d-orbitals.

In a 'low spin' state, the lower energy d-orbitals are filled first, by pairing up if necessary, before the higher energy d-orbitals are used.

Use diagrams like the one above to show the electronic distribution of a Fe^{2+} ion in a high spin state, and in a low spin state.

[2]

(v) State and explain which ligand will result in a larger energy gap, E, between its d-orbitals.

[1]

[Total: 20]

5 Carbonyl compounds are common in our everyday lives. They are mainly used as solvents, perfumes and flavouring agents or as intermediates in the manufacture of plastics and pharmaceuticals.

(a) The characteristic smell of cherries and fresh almonds is due to benzaldehyde.

(i) Benzaldehyde can react with chloromethane to form 3-methylbenzaldehyde. Describe the mechanism for this reaction.

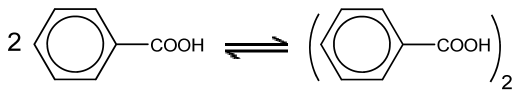
[3]

(ii) Benzene can also react under a similar reaction with chloromethane.

State and explain whether benzene or benzaldehyde would react with chloromethane more readily.

[2]

(b) Benzaldehyde can also react with hot acidified dichromate(VI) to give benzoic acid. In benzene, benzoic acid associates to form dimers.



(i) Draw a diagram to illustrate the bonding in the dimer.

[1]

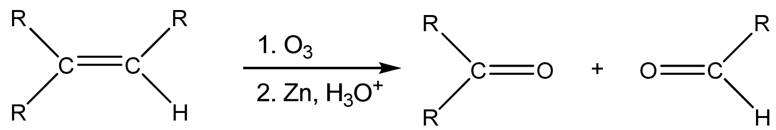
(ii) Predict and explain whether the dimerisation is favoured at a high or low temperature.

[2]

(iii) Suggest why the above equilibrium cannot be established in an aqueous medium.

[1]

(c) Carbonyl compounds can be prepared from alkenes via the ozonolysis reaction as shown below.



An alkene **M**, $C_{11}H_{14}$ was treated with O_3, followed by Zn and H_3O^+ to give **N** C_4H_8O and **P**, C_7H_6O. **N** gives a yellow precipitate with aqueous alkaline iodine while **P** gives a grey precipitate with ammonical silver nitrate solution. In the presence of OH^-, **P** undergoes a reaction to give **Q**, $C_7H_6O_2$ and **S**, C_7H_8O. Both **Q** and **S** react with Na metal, but only **Q** reacts with $NaHCO_3$.

Suggest the structures of **M, N, P, Q** and **S**.

[5]

(d) **T, U, V** and **W** are four consecutive elements in the **fourth** period of the Periodic Table. (The letters are **not** the actual symbols of the elements.)

T is a soft, silvery metal with a melting point just above room temperature. Its amphoteric oxide, $\mathbf{T}_2O_3$, has a melting point of 1900 °C and can be formed by heating **T** in oxygen.

W is a solid that can exist as several allotropes, most of which contain $\mathbf{W}_8$ molecules. **W** burns in air to form $\mathbf{W}O_2$ and $\mathbf{W}O_3$**,** which dissolves in water to form an acidic solution. The acidic solutions react with sodium hydroxide to form the salt $Na_2\mathbf{W}O_3$ and $Na_2\mathbf{W}O_4$ respectively.

(i) Suggest the identities of **T** and **W**.

[2]

(ii) Write equations for the reactions of $\mathbf{T}_2O_3$ with
- hydrochloric acid,
- sodium hydroxide

[2]

(iii) Suggest the structure in $\mathbf{T}_2O_3$.

[1]

(iv) Write an equation for the formation of the acidic solution when $\mathbf{W}O_3$ dissolves in water.

[1]

[Total: 20]

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